

L3 | Real Project Records: Interfaces, Assumptions, and the Visibility of Failure

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L3 is the layer of this series dedicated to real project narratives.

Unlike L1, which establishes methodology, and L2, which develops abstract judgment models, L3 focuses on concrete events from real projects. Through verifiable processes and outcomes, it examines how systems, procedures, and professional judgment shift under real-world conditions.

L3 does not aim to provide standard answers, nor does it serve as a retrospective of success.

Its focus is on issues that are tacitly accepted, postponed, or assumed to be “still manageable,” and how such issues gradually alter system premises and ultimately shape project outcomes.

Within L3, each chapter addresses one category of problem.

The narrative avoids dramatization and instead emphasizes stages, interfaces, decisions, and consequences.

These records are neither incident reports nor best-practice summaries, but real samples intended to expose blind spots in judgment.

L3-1 | The Problem of Grey Zones

In engineering and system projects, the causes of long-term loss are rarely explicit errors. More often, they are issues that were never clearly asked. These issues do not violate codes, nor do they appear unreasonable. They exist quietly between processes, being “reasonably” overlooked across disciplines and project phases.

Such issues typically emerge at interfaces.

A change does not negate an existing design, yet subtly alters its underlying assumptions.

A requirement appears to remain within the original scope, yet loses ownership at the boundary of responsibility.

Because these issues do not immediately trigger mandatory reviews or visible failures, they are often deferred as matters to be addressed later. I refer to these issues as grey zones.

They are not technical blind spots, but procedural ones.

Not a lack of competence, but a lack of obligation to respond.

L3-1 focuses on how such issues manifest in real projects.

The following case is not an exception, but one of the most common and easily overlooked forms of grey zones.

Definition of Grey Zones (Working Definition)

In this series, a “grey zone” does not refer to a technical challenge or a code conflict. It refers to a category of issues that simultaneously meet the following conditions:

- The issue objectively exists, but has not been formally taken over by any discipline.
 - The issue does not cause immediate failure and therefore does not trigger mandatory processes or reviews.
 - The issue affects system assumptions, but this impact has not been explicitly recognized.
 - Responsibility boundaries are blurred across disciplines or project phases.
 - Each party's judgment is internally reasonable, yet no closed loop is formed.
- When these conditions coexist, an issue naturally slips into a grey zone.

It is neither rejected nor resolved, but continuously deferred until the optimal window for intervention is lost.

L3-1 examines how this category of issues appears in real project settings. The following case illustrates one of the most typical grey-zone scenarios.

L3-1-1 | Slab Penetrations in Precast Floors and the Missing Structural Calculation

Project Background (Factual Summary)

- Project year: 2018
- Location: Shanghai, China
- Project type: Foreign-invested laboratory campus
- Building type: Standalone laboratory building
- Functional composition: Laboratory spaces combined with administrative (non-lab) areas
- Project phase: Schematic / preliminary design
- Current milestone: Structural design input stage; construction drawings not yet commenced
- My scope of work: Interface coordination within laboratory partition and internal systems
- System characteristics: Laboratory areas planned with modular integrated support (combined service racks) and highly integrated MEP and process services
- Potential impact: Increased requirements for early slab penetration planning and structural input, not yet formally translated into structural design parameters

Triggering Event

During the structural design input stage, a slab penetration requirement emerged that appeared reasonable on the surface.

The project adopted a precast slab system, and based on prevailing workflow assumptions, this stage was still considered adjustable. The issue was therefore treated as something that could be coordinated later.

Two default assumptions emerged simultaneously:

- The penetration requirement originated from internal laboratory system layouts and therefore belonged to our system scope.
- The penetration directly affected the structural system and therefore fell under the responsibility of the structural engineer.

Both assumptions were internally reasonable within their respective frameworks.

Their combined outcome, however, was that no structural calculation was submitted.

Risk Identification

What triggered concern was not the penetration itself, but the convergence of several signals:

- Structural calculations were repeatedly mentioned in internal discussions but never formally entered the design input process.
- Judgments remained confined to internal discussions and failed to form a cross-disciplinary loop.
- External coordination was conducted primarily between individual systems and the client, without synchronized engagement with the lead structural designer.

This condition implied that once key milestones were passed, any structural implications of slab penetrations would fall into a state with no accountable owner.

Intervention

Rather than assigning fault, I raised three questions that required explicit answers:

- Ownership and confirmation workflow for slab penetration requirements.
- Penetration dimensions, elevations, and reference coordinates.
- Responsibility for issuing structural calculations and whether impacts existed on reinforcement, fire zoning, or laboratory cleanliness.

I then formally raised the issue to the client, requesting clarification of scope and initiating coordinated discussions among structural, MEP, and construction stakeholders. In parallel:

- A penetration register was established.
- Key penetration nodes were frozen.
- Default assumptions were converted into explicit technical inputs through RFIs and issue logs.

Outcome

Once the risks were made explicit, the project entered a realignment phase.

Structural, MEP, and related disciplines conducted focused coordination, consolidating previously fragmented judgments into a unified technical context.

Final alignment included:

- Formal classification of slab penetrations as early structural design inputs.
- Completion of structural calculations by the lead structural consultant.
- Adjustment of penetration locations and dimensions.
- Targeted optimization of internal laboratory system layouts to avoid passive modifications to the precast slab system.

This intervention did not result in noticeable schedule impact. Instead, it resolved potential risks within the remaining structural flexibility window. In subsequent construction drawing stages, the issue did not reappear as a variation or rework item, allowing the project to proceed within controlled parameters.

L3-1-1 | Phase Judgment

When evaluated against the earlier definition, the grey-zone characteristics of this case become evident.

The penetration requirement objectively existed yet was not formally owned.

Its impacts were delayed and therefore failed to trigger mandatory reviews.

Responsibility oscillated across disciplines without closure.
Slab penetrations themselves are not complex, nor are they difficult to calculate.

What creates complexity is their dual status in the process: treated as “still adjustable” while already altering the assumptions underpinning the structural system.

This represents the first category of grey-zone issues addressed in L3. Subsequent chapters will examine additional grey-zone patterns under different project conditions, before consolidating them into a unified judgment framework at the conclusion of L3.

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